

## 33.3 Repeated measures

A repeated-measures study observes the same subject on several occasions: a clustered design whose cluster is the subject and whose “treatments” are the occasions, laid out like the split-plot design of [Section 33.1](#), and for many years analysed that way.

The analogy has one weak joint. What makes the split-plot covariance compound symmetric is the *randomization* of subplot treatments within the whole plot, and occasions cannot be randomized: measurements close together in time are more alike than distant ones, so compound symmetry here is usually false. What is remarkable is that the univariate  $F$  test survives a much weaker condition, and that the condition can be written down exactly.

### 33.3.1 The univariate analysis

Let  $\mathbf{y}_1, \dots, \mathbf{y}_N$  be independent  $N_m(\boldsymbol{\mu}, \boldsymbol{\Sigma})$  vectors, one per subject, with  $\boldsymbol{\Sigma}$  positive definite and unknown, and write  $y_{ij}$  for occasion  $j$  of subject  $i$ . The hypothesis of interest is

$$H : \mu_1 = \dots = \mu_m, \quad \text{that is,} \quad \boldsymbol{\mu} \in \text{span}\{\mathbf{1}\}.$$

Laid out with subjects as rows and occasions as columns the data have one observation per cell, and the classical analysis is the additive two-way analysis of variance of [Section 16.1](#), with the subject-by-occasion interaction as error. With  $\bar{y}_{i\cdot}$ ,  $\bar{y}_{\cdot j}$  and  $\bar{y}$  the subject, occasion and grand means,

$$SS_T = N \sum_{j=1}^m (\bar{y}_{\cdot j} - \bar{y})^2, \quad SS_E = \sum_{i=1}^N \sum_{j=1}^m (y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y})^2, \quad (33.3.1)$$

and the statistic is  $F = \{SS_T/(m-1)\}/\{SS_E/((N-1)(m-1))\}$ . It is the subplot stratum of a split-plot design with one whole plot per subject, and under compound symmetry it is exact ([Theorem 33.3.2\(b\)](#) below, with [Exercise \(B1\)](#)). The question is how much of that exactness survives a general  $\boldsymbol{\Sigma}$ .

Everything reduces to the within-subject contrasts. Let  $\mathbf{U}$  be  $m \times (m-1)$  with orthonormal columns orthogonal to  $\mathbf{1}$ , so that  $\mathbf{U}\mathbf{U}^\top = \mathbf{I} - \bar{\mathbf{J}}_m$ , and set  $\mathbf{z}_i = \mathbf{U}^\top \mathbf{y}_i$ : independent  $N_{m-1}(\mathbf{U}^\top \boldsymbol{\mu}, \boldsymbol{\Psi})$  with  $\boldsymbol{\Psi} = \mathbf{U}^\top \boldsymbol{\Sigma} \mathbf{U}$ , with  $H$  saying exactly that  $\mathbf{U}^\top \boldsymbol{\mu} = \mathbf{0}$ . The subject means carry no information about  $H$ .

### 33.3.2 Sphericity

#### Definition 33.3.1. Sphericity and Box's measure

Let  $\boldsymbol{\Sigma}$  be an  $m \times m$  positive definite matrix and  $\mathbf{U}$  as above.  $\boldsymbol{\Sigma}$  satisfies **sphericity** if  $\mathbf{U}^\top \boldsymbol{\Sigma} \mathbf{U} = \lambda \mathbf{I}_{m-1}$  for some  $\lambda > 0$ . **Box's measure of sphericity** is

$$\epsilon(\boldsymbol{\Sigma}) = \frac{\{\text{tr } \boldsymbol{\Psi}\}^2}{(m-1) \text{tr}(\boldsymbol{\Psi}^2)}, \quad \boldsymbol{\Psi} = \mathbf{U}^\top \boldsymbol{\Sigma} \mathbf{U}. \quad (33.3.2)$$

Neither the condition nor  $\epsilon$  depends on the choice of  $\mathbf{U}$ : two such matrices satisfy  $\tilde{\mathbf{U}} = \mathbf{U}\mathbf{Q}$

with  $\mathbf{Q}$  orthogonal, so  $\tilde{\Psi} = \mathbf{Q}^\top \Psi \mathbf{Q}$  has the same trace, the same trace of square, and is a multiple of the identity exactly when  $\Psi$  is.

### Lemma 33.3.1. Equivalent forms of sphericity

For a positive definite  $m \times m$  matrix  $\Sigma$  the following are equivalent:

- (i)  $\mathbf{U}^\top \Sigma \mathbf{U} = \lambda \mathbf{I}_{m-1}$ ;
- (ii)  $\text{Var}(Y_j - Y_k) = 2\lambda$  for every  $j \neq k$ ;
- (iii)  $\sigma_{jk} = a_j + a_k + \lambda \delta_{jk}$  for some  $a_1, \dots, a_m$ , where  $\delta_{jk}$  is one if  $j = k$  and zero otherwise;
- (iv)  $\Sigma = \lambda \mathbf{I} + \mathbf{1}\mathbf{a}^\top + \mathbf{a}\mathbf{1}^\top$  for some  $\mathbf{a} \in \mathbb{R}^m$ .

Moreover  $1/(m-1) \leq \epsilon(\Sigma) \leq 1$ , with  $\epsilon = 1$  iff sphericity holds, and compound symmetry implies sphericity but not conversely.

*Proof.* Write  $\mathbf{C} = \mathbf{I} - \bar{\mathbf{J}}_m = \mathbf{U}\mathbf{U}^\top$ . Since  $\mathbf{U}^\top \mathbf{U} = \mathbf{I}_{m-1}$ , condition (i) holds iff  $\mathbf{C}\Sigma\mathbf{C} = \lambda\mathbf{C}$ .

- (i)  $\Rightarrow$  (ii):  $\mathbf{e}_j - \mathbf{e}_k$  is orthogonal to  $\mathbf{1}$ , hence fixed by  $\mathbf{C}$ , so  $\text{Var}(Y_j - Y_k)$  equals  $(\mathbf{e}_j - \mathbf{e}_k)^\top \mathbf{C}\Sigma\mathbf{C}(\mathbf{e}_j - \mathbf{e}_k)$ , which is  $\lambda\|\mathbf{e}_j - \mathbf{e}_k\|^2 = 2\lambda$ .
- (ii)  $\Rightarrow$  (iii):  $\text{Var}(Y_j - Y_k) = \sigma_{jj} + \sigma_{kk} - 2\sigma_{jk} = 2\lambda$  gives  $\sigma_{jk} = (\sigma_{jj} - \lambda)/2 + (\sigma_{kk} - \lambda)/2$  for  $j \neq k$ . Put  $a_j = (\sigma_{jj} - \lambda)/2$ ; then  $\sigma_{jk} = a_j + a_k$  off the diagonal and  $\sigma_{jj} = 2a_j + \lambda$  on it, which is (iii).
- (iii)  $\Leftrightarrow$  (iv) is the same statement in matrix form.
- (iv)  $\Rightarrow$  (i):  $\mathbf{C}\mathbf{1} = \mathbf{0}$ , so  $\mathbf{C}\Sigma\mathbf{C} = \lambda\mathbf{C}$ .

For the bounds, let  $\psi_1, \dots, \psi_{m-1} > 0$  be the eigenvalues of  $\Psi$ . Then  $\epsilon = (\sum_l \psi_l)^2 / \{(m-1) \sum_l \psi_l^2\}$ . The Cauchy–Schwarz inequality gives  $(\sum_l \psi_l)^2 \leq (m-1) \sum_l \psi_l^2$  with equality iff all  $\psi_l$  are equal, so  $\epsilon \leq 1$  with equality iff  $\Psi = \lambda \mathbf{I}$ ; and  $\sum_l \psi_l^2 \leq (\sum_l \psi_l)^2$  for nonnegative numbers gives  $\epsilon \geq 1/(m-1)$ . Compound symmetry is (iv) with all  $a_j$  equal; taking the  $a_j$  unequal gives sphericity without compound symmetry, and such a  $\Sigma$  is positive definite as soon as  $\lambda$  is large enough. ■

Condition (ii) is the form to remember and to check: *all pairwise differences of the repeated measurements have the same variance*. Compound symmetry says more than the  $F$  test needs.

### Theorem 33.3.2. Sphericity is necessary and sufficient for the exact $F$ test

Let  $\mathbf{y}_1, \dots, \mathbf{y}_N$  be independent  $N_m(\boldsymbol{\mu}, \Sigma)$  with  $\Sigma$  positive definite,  $m \geq 2$ ,  $N \geq 2$ , and let  $F$  be the statistic built from Eq. 33.3.1. Put  $\Psi = \mathbf{U}^\top \Sigma \mathbf{U}$ ,  $\mathbf{q} = \sqrt{N} \bar{\mathbf{z}}$  and  $\mathbf{W} = \sum_i (\mathbf{z}_i - \bar{\mathbf{z}})(\mathbf{z}_i - \bar{\mathbf{z}})^\top$ .

- (a)  $\text{SS}_T = \|\mathbf{q}\|^2$  and  $\text{SS}_E = \text{tr}(\mathbf{W})$ , and under  $H$  the vector  $\mathbf{q} \sim N_{m-1}(\mathbf{0}, \Psi)$  is independent of  $\mathbf{W} \sim W_{m-1}(N-1, \Psi)$ . Hence  $F = N(N-1)\|\bar{\mathbf{z}}\|^2 / \text{tr}(\mathbf{W})$ .
- (b) If  $\Sigma$  satisfies sphericity, then under  $H$   $F \sim F(m-1, (N-1)(m-1))$  exactly, for every  $N \geq 2$ .
- (c) Conversely, if the null distribution of  $F$  is  $F(m-1, (N-1)(m-1))$  for every  $N \geq 2$ , then  $\Sigma$  satisfies sphericity.

*Proof.*

- (a) Since  $\mathbf{C} = \mathbf{U}\mathbf{U}^\top$  and  $\mathbf{U}$  has orthonormal columns,  $\sum_j (\bar{y}_{\cdot j} - \bar{y})^2 = \|\mathbf{C}\bar{\mathbf{y}}\|^2 = \|\bar{\mathbf{z}}\|^2$ , so  $\text{SS}_T = N\|\bar{\mathbf{z}}\|^2 = \|\mathbf{q}\|^2$ ; likewise subject  $i$ 's contribution to  $\text{SS}_E$  is  $\|\mathbf{z}_i - \bar{\mathbf{z}}\|^2$ , summing to  $\text{tr}(\mathbf{W})$ . Under  $H$  the  $\mathbf{z}_i$  are independent  $N_{m-1}(\mathbf{0}, \Psi)$ , so Lemma 14.2.3 gives the Wishart law of  $\mathbf{W}$  and its independence of  $\bar{\mathbf{z}}$ . Finally  $F = \{\|\mathbf{q}\|^2/(m-1)\} / \{(N-1)(m-1)/\text{tr} \mathbf{W}\}$ .
- (b) If  $\Psi = \lambda \mathbf{I}_{m-1}$ , then  $\mathbf{q}/\sqrt{\lambda}$  is standard normal, so  $\|\mathbf{q}\|^2/\lambda \sim \chi^2(m-1)$ . And  $\mathbf{W}/\lambda \sim W_{m-1}(N-1, \mathbf{I})$ , which is the matrix of sums of squares and cross products of  $(N-1)(m-1)$  independent standard normal variables, so  $\text{tr}(\mathbf{W})/\lambda \sim \chi^2((N-1)(m-1))$ . The two are independent, and  $\lambda$  cancels from the ratio, leaving the ratio of two independent chi-squared variables each divided by its degrees of freedom (Definition 4.1.2).
- (c) Write  $F_N$  for the statistic with  $N$  subjects. By (a),

$$F_N = \|\mathbf{q}\|^2 \cdot \frac{N-1}{\text{tr}(\mathbf{W})},$$

where  $\mathbf{q} \sim N_{m-1}(\mathbf{0}, \Psi)$  has the *same* distribution for every  $N$  and is independent of  $\mathbf{W}$ . Since  $\mathbf{W}/(N-1)$  is an average of  $N-1$  independent  $W_{m-1}(1, \Psi)$  matrices, the weak law of large numbers gives  $\text{tr}(\mathbf{W})/(N-1) \rightarrow \text{tr} \Psi$  in probability, so by Slutsky's lemma  $F_N \rightarrow \|\mathbf{q}\|^2 / \text{tr} \Psi$  in distribution. On the other hand  $F(m-1, \nu) \rightarrow \chi^2(m-1)/(m-1)$  in distribution as  $\nu \rightarrow \infty$ , because its denominator chi-squared divided by  $\nu$  tends to one in probability. If the exactness holds for every  $N$ , the two limits agree:

$$\|\mathbf{q}\|^2 \stackrel{d}{=} \frac{\text{tr} \Psi}{m-1} \chi^2(m-1).$$

Now match variances. By Theorem 4.2.3,  $\text{Var}(\|\mathbf{q}\|^2) = 2 \text{tr}(\Psi^2)$ , while the right-hand side has variance  $2\{\text{tr} \Psi\}^2/(m-1)$ . Equality says  $(m-1) \text{tr}(\Psi^2) = \{\text{tr} \Psi\}^2$ , that is  $\epsilon(\Sigma) = 1$ , and Lemma 33.3.1 makes that sphericity. ■

*Idea.* The repeated-measures  $F$  test does not require the occasions to be exchangeable, only that the  $m-1$  orthonormalized within-subject contrasts be uncorrelated with equal variances — that the contrast space look spherical, which series of measurements in time almost never do.

### 33.3.3 Two corrections and one alternative

When sphericity fails,  $\|\mathbf{q}\|^2$  and  $\text{tr}(\mathbf{W})$  are still weighted sums of chi-squared variables, and matching their first two moments to scaled chi-squared variables gives a usable approximation (Box 1954).

#### Proposition 33.3.3. The moment-matched approximation

Under the assumptions of Theorem 33.3.2, with  $H$  true, write  $\epsilon = \epsilon(\Sigma)$ ,  $d_1 = m-1$  and  $d_2 = (N-1)(m-1)$ . Then

- (a)  $E SS_T = \text{tr } \Psi$ ,  $\text{Var } SS_T = 2 \text{tr}(\Psi^2)$ ,  $E SS_E = (N - 1) \text{tr } \Psi$  and  $\text{Var } SS_E = 2(N - 1) \text{tr}(\Psi^2)$ .
- (b) The scaled chi-squared variable  $g\chi^2(h)$  matching the first two moments of  $SS_T$  has  $h = \epsilon d_1$  and  $g = \text{tr}(\Psi^2)/\text{tr } \Psi$ ; the one matching  $SS_E$  has  $h' = \epsilon d_2$  and the *same*  $g$ . Since  $SS_T$  and  $SS_E$  are independent by [Theorem 33.3.2\(a\)](#), replacing each by its matched scaled chi-squared gives the approximation  $F \approx F(\epsilon d_1, \epsilon d_2)$ , which is exact when  $\epsilon = 1$ .

*Proof.*

- (a) The first two are Theorem 4.2.3 with  $\mathbf{A} = \mathbf{I}$  and mean zero. Writing  $\mathbf{W} = \sum_{l=1}^{N-1} \mathbf{w}_l \mathbf{w}_l^\top$  with  $\mathbf{w}_l$  independent  $N_{m-1}(\mathbf{0}, \Psi)$  (Lemma 14.2.3),  $\text{tr } \mathbf{W} = \sum_l \|\mathbf{w}_l\|^2$  is a sum of  $N - 1$  independent copies of  $\|\mathbf{q}\|^2$ , so its mean and variance are  $N - 1$  times theirs.
- (b) Matching  $gh = \text{tr } \Psi$  and  $2g^2h = 2 \text{tr}(\Psi^2)$  gives  $g = \text{tr}(\Psi^2)/\text{tr } \Psi$  and  $h = \{\text{tr } \Psi\}^2/\text{tr}(\Psi^2) = \epsilon d_1$ . For  $SS_E$  both moments are multiplied by  $N - 1$ , so  $g$  is unchanged and  $h' = (N - 1)h = \epsilon d_2$ . Since  $d_2/d_1 = h'/h$ , the ratio  $(SS_T/d_1)/(SS_E/d_2)$  becomes  $\{\chi^2(h)/h\}/\{\chi^2(h')/h'\}$ , an  $F(h, h')$  variable, the two chi-squared variables being independent by [Theorem 33.3.2\(a\)](#); at  $\epsilon = 1$  this is the exact result of [Theorem 33.3.2\(b\)](#). ■

By [Lemma 33.3.1](#),  $\epsilon \geq 1/(m - 1)$ , so the approximating distribution is never more dispersed than  $F(d_1/(m - 1), d_2/(m - 1)) = F(1, N - 1)$ , the largest that [Proposition 33.3.3\(b\)](#) can ask for. Referring  $F$  to  $F(1, N - 1)$  is therefore conservative within Box's approximation, and Geisser and Greenhouse (1958) proved it exactly conservative, for every  $\Sigma$  and  $N$ . This is the **lower-bound** correction, turned into a recommendation by Greenhouse and Geisser (1959).

Part (b) becomes a procedure once  $\epsilon$  is estimated from the sample covariance matrix  $\mathbf{S}$  of the  $\mathbf{y}_i$ .

- The **Greenhouse–Geisser** correction refers  $F$  to  $F(\hat{\epsilon}d_1, \hat{\epsilon}d_2)$  with the plug-in  $\hat{\epsilon} = \epsilon(\mathbf{S})$ , which is biased downward, so the test is conservative.
- The **Huynh–Feldt** correction removes the leading bias. For a single group of  $N$  subjects it uses

$$\tilde{\epsilon} = \min\left\{1, \frac{N(m - 1)\hat{\epsilon} - 2}{(m - 1)\{(N - 1) - (m - 1)\hat{\epsilon}\}}\right\}, \quad (33.3.3)$$

which exceeds  $\hat{\epsilon}$  and gives a less conservative test; its own bias is upward when  $\epsilon$  is small.

- The **lower-bound** correction takes  $\epsilon = 1/(m - 1)$ : safe, and usually too conservative to be useful.

A third option avoids the issue. Since the  $\mathbf{z}_i$  are independent  $N_{m-1}(\mathbf{U}^\top \boldsymbol{\mu}, \Psi)$  with  $\Psi$  unrestricted,  $H$  says a multivariate normal mean is zero, and the classical test is Hotelling's.

#### Proposition 33.3.4. The multivariate test

Under the assumptions of [Theorem 33.3.2](#) with  $N \geq m$ , let  $\mathbf{S}_z = \mathbf{W}/(N - 1)$  and

$$T^2 = N \bar{\mathbf{z}}^\top \mathbf{S}_z^{-1} \bar{\mathbf{z}}.$$

Then  $T^2$  does not depend on the choice of  $\mathbf{U}$ , nor on replacing  $\mathbf{U}$  by  $\mathbf{UK}$  for any nonsingular  $(m-1) \times (m-1)$  matrix  $\mathbf{K}$ ; and under  $H$ ,

$$\frac{N-m+1}{(N-1)(m-1)} T^2 \sim F(m-1, N-m+1),$$

whatever  $\Sigma$  is. The distributional statement is Hotelling's and is not proved here; see Anderson (2003, §5.2).

*Proof.* Only the invariance is proved. Replacing  $\mathbf{U}$  by  $\mathbf{UK}$  replaces  $\bar{\mathbf{z}}$  by  $\mathbf{K}^\top \bar{\mathbf{z}}$  and  $\mathbf{S}_z$  by  $\mathbf{K}^\top \mathbf{S}_z \mathbf{K}$ , leaving  $T^2$  unchanged; two orthonormal bases differ by an orthogonal  $\mathbf{K}$ . The requirement  $N \geq m$  makes  $\mathbf{S}_z$ , a Wishart matrix on  $N-1 \geq m-1$  degrees of freedom, invertible with probability one. ■

Exactness under any  $\Sigma$  sounds decisive, but it costs power: the multivariate test estimates a whole covariance matrix and has  $N-m+1$  denominator degrees of freedom rather than  $(N-1)(m-1)$ . When sphericity nearly holds the univariate test is much more powerful, when it fails badly the two are comparable, and when  $N < m$  the multivariate test does not exist.

### 33.3.4 An example, four ways

#### Example. Grip strength over six visits

Twenty-four patients have grip strength measured, in kilograms, at six weekly visits, simulated from  $N_6(\boldsymbol{\mu}, \Sigma)$  with  $\Sigma$  first-order autoregressive,  $\sigma_{jk} = 4 \times 0.6^{|j-k|}$ , and a mean drifting slowly upward; the observed occasion means are 47.818, 48.099, 49.056, 49.177, 49.561 and 49.263 kg. The true Box measure is  $\epsilon = 0.700$ : adjacent visits correlate 0.6 and visits five weeks apart only  $0.6^5$ , so adjacent differences are much less variable than distant ones. The four analyses of  $H$  give

| Analysis                                        | statistic | degrees of freedom | $p$    |
|-------------------------------------------------|-----------|--------------------|--------|
| univariate,<br>uncorrected                      | 4.770     | 5, 115             | 0.0005 |
| Greenhouse–Geisser,<br>$\hat{\epsilon} = 0.645$ | 4.770     | 3.23, 74.2         | 0.0035 |
| Huynh–Feldt,<br>$\tilde{\epsilon} = 0.763$      | 4.770     | 3.81, 87.7         | 0.0019 |
| multivariate                                    | 3.605     | 5, 19              | 0.0184 |

All four reject, but the evidence differs by a factor of nearly forty between first and last. Distrust the uncorrected test: its nominal degrees of freedom are too large by the factor  $\epsilon$ .

```
import numpy as np
from scipy import stats
```

```
N, m = 24, 6
```

```

mu_true = np.array([48.0, 48.5, 48.9, 49.2, 49.4, 49.5]) # kg, a slow improvement
sigma, rho_true = 2.0, 0.6
lags = np.abs(np.subtract.outer(np.arange(m), np.arange(m)))
Sigma_true = sigma**2 * rho_true**lags # AR(1) within a patient
rng = np.random.default_rng(3303)
Yobs = np.round(mu_true + rng.multivariate_normal(np.zeros(m), Sigma_true, N), 2)

def contrast_basis(m):
    """Orthonormal basis U of the space of contrasts among m repeated measures."""
    Q, _ = np.linalg.qr(np.column_stack([np.ones(m), np.eye(m)[:, 1:]])
    return Q[:, 1:] # columns orthonormal and
                    orthogonal to 1

def epsilon(Psi):
    """Box's measure of departure from sphericity, 1/(m-1) <= eps <= 1."""
    k = Psi.shape[0]
    return np.trace(Psi)**2 / (k * np.sum(Psi * Psi))

U = contrast_basis(m)
Z = Yobs @ U # N x (m-1) contrasts, one row
              per patient
zbar = Z.mean(axis=0)
Sz = np.cov(Z, rowvar=False, ddof=1) # Wishart, (N-1) degrees of
                                     freedom

F_uncorrected = N * (zbar @ zbar) / np.trace(Sz)
eps_hat = epsilon(Sz)
eps_hf = min(1.0, (N * (m - 1) * eps_hat - 2)
             / ((m - 1) * ((N - 1) - (m - 1) * eps_hat)))
d1, d2 = m - 1, (N - 1) * (m - 1)
p_un = stats.f.sf(F_uncorrected, d1, d2)
p_gg = stats.f.sf(F_uncorrected, d1 * eps_hat, d2 * eps_hat)
p_hf = stats.f.sf(F_uncorrected, d1 * eps_hf, d2 * eps_hf)
T2 = N * zbar @ np.linalg.solve(Sz, zbar) # Hotelling's statistic
F_mv = (N - m + 1) / ((N - 1) * (m - 1)) * T2
p_mv = stats.f.sf(F_mv, m - 1, N - m + 1)
print(f"F = {F_uncorrected:.3f} on ({d1}, {d2}) df, p = {p_un:.4f}")
print(f"Greenhouse-Geisser eps = {eps_hat:.3f}, p = {p_gg:.4f}")
print(f"Huynh-Feldt eps = {eps_hf:.3f}, p = {p_hf:.4f}")
print(f"multivariate F({m - 1}, {N - m + 1}) = {F_mv:.3f}, p = {p_mv:.4f}")

```

[Run this code online](#)

How much does the failure of sphericity cost? [Figure 33.3.1](#) answers with a simulation under  $H$ . Panel (a) shows the null distribution of the uncorrected  $F$ : more dispersed than  $F(5, 115)$ , with a heavier right tail, and tracking [Proposition 33.3.3](#) closely. Panel (b) traces the level of a nominal 5% test as the autoregressive parameter grows. At  $\rho = 0$  it is 0.050, as [Theorem 33.3.2\(b\)](#) requires; by  $\rho = 0.8$ , where  $\epsilon = 0.562$ , it is 0.085. The Greenhouse–Geisser test is conservative throughout (0.038 to 0.049), the Huynh–Feldt test close to nominal (0.049 to 0.056), and the multivariate test exact at 0.049 for every  $\rho$ . A spherical covariance of the form [Lemma 33.3.1\(iv\)](#) with unequal  $a_j$ , far from compound symmetric, gives the uncorrected test a level of 0.050: sphericity, not compound symmetry, is what matters.

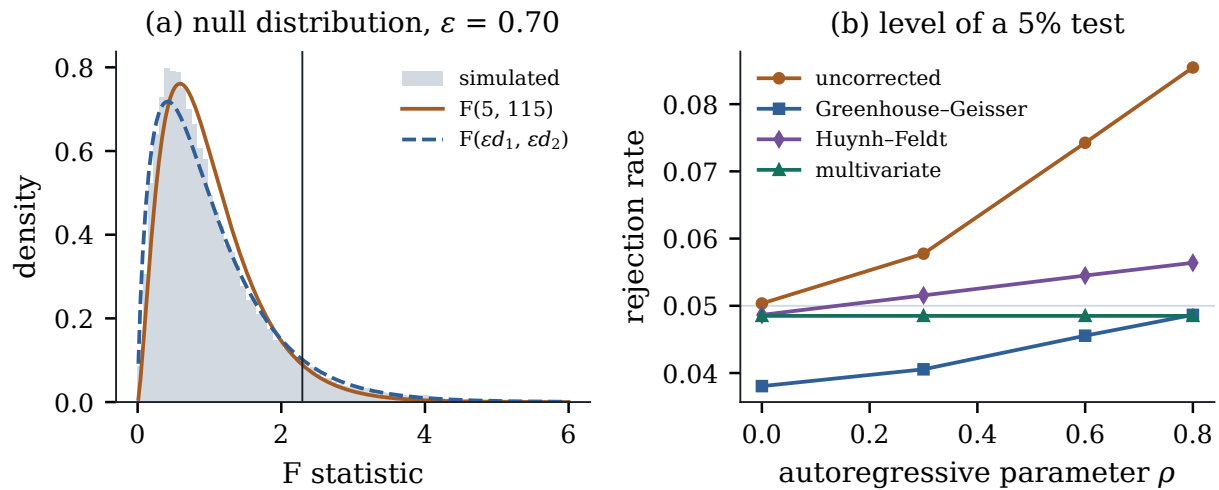


Figure 33.3.1: (a) The null distribution of the uncorrected  $F$  statistic with autoregressive errors, against the nominal  $F(5, 115)$  density and the moment-matched  $F(\epsilon d_1, \epsilon d_2)$  of Proposition 33.3.3, with the nominal 5% point marked. (b) Simulated level of four tests as  $\rho$  grows.

### 33.3.5 The modern treatment

The corrections above patch a fixed analysis; the mixed-model approach fits the covariance instead. Write the data as a linear mixed model (Definition 32.2.1) with a fixed effect for each occasion and a within-subject covariance  $\mathbf{R}$  of chosen shape, estimate its parameters by restricted maximum likelihood (Definition 32.4.1), and test the fixed effects by a Wald statistic with estimated degrees of freedom (Proposition 32.5.1). Compound symmetric  $\mathbf{R}$  reproduces the uncorrected univariate  $F$  (Exercise (B2)), unstructured  $\mathbf{R}$  the multivariate test up to the degrees-of-freedom approximation, and anything between is a model the classical scheme has no room for. Unbalanced data and missing visits are handled too, whereas Eq. 33.3.1 and  $T^2$  need a complete rectangle. The price is that the covariance must be chosen, and the choice affects the standard errors: Section 33.4.

*Remark* (Between-subjects factors). With a between-subjects factor the within-subject analysis is the same, applied to the pooled within-group covariance, and the univariate tests are exact iff the pooled  $\Psi$  is spherical *and* the groups share a common  $\Sigma$ . Mauchly's (1940) test of sphericity itself, which most software reports, is sensitive to nonnormality and, like any preliminary test, distorts the level of what follows.

### 33.3.6 Exercises

#### 33.3.6.1 A. Check your understanding

##### Exercise. A1

Show that every  $2 \times 2$  covariance matrix satisfies sphericity, so that with  $m = 2$  the repeated-measures  $F$  test is exact whatever  $\Sigma$  is. Identify the test in that case.

*Solution.* With  $m = 2$ ,  $\Psi = \mathbf{U}^\top \Sigma \mathbf{U}$  is  $1 \times 1$ , hence a multiple of  $\mathbf{I}_1$ . By Theorem 33.3.2(b) the test is exact. It is the paired  $t$  test: the single contrast is  $(y_{i1} - y_{i2})/\sqrt{2}$ , and  $F$  is the

square of the paired  $t$  statistic on  $N - 1$  degrees of freedom.

### Exercise. A2

For  $m = 3$  and an autoregressive covariance  $\sigma_{jk} = \rho^{|j-k|}$  with  $0 < \rho < 1$ , compute  $\Psi$  for the basis  $\mathbf{u}_1 = (1, -1, 0)^\top / \sqrt{2}$ ,  $\mathbf{u}_2 = (1, 1, -2)^\top / \sqrt{6}$  and show that sphericity fails. Which pairwise difference has the largest variance?

### 33.3.6.2 B. Practice

### Exercise. B1

Show directly that compound symmetry  $\Sigma = \sigma^2\{(1 - \rho)\mathbf{I} + \rho\mathbf{1}\mathbf{1}^\top\}$  gives  $\Psi = \sigma^2(1 - \rho)\mathbf{I}_{m-1}$ , and hence that the split-plot analysis of Theorem 33.1.2 is the repeated-measures  $F$  test with  $\epsilon = 1$ . Which  $\lambda$  of Lemma 33.3.1 does it correspond to?

*Solution.*  $\mathbf{U}^\top \mathbf{1} = \mathbf{0}$ , so  $\Psi = \mathbf{U}^\top \Sigma \mathbf{U} = \sigma^2(1 - \rho)\mathbf{U}^\top \mathbf{U} = \sigma^2(1 - \rho)\mathbf{I}_{m-1}$ . Hence  $\epsilon = 1$  and  $\lambda = \sigma^2(1 - \rho)$ , which is the subplot stratum eigenvalue  $\lambda_s$  of Lemma 33.1.1. In Lemma 33.3.1(iv) take  $\mathbf{a} = (\sigma^2\rho/2)\mathbf{1}$ .

### Exercise. B2

Consider the mixed model  $y_{ij} = \mu_j + u_i + e_{ij}$  with  $u_i$  and  $e_{ij}$  independent normal, of variances  $\sigma_u^2$  and  $\sigma_e^2$ . Show that restricted maximum likelihood gives  $\hat{\sigma}_e^2 = \text{MS}_E$  and  $\hat{\sigma}_u^2 = (\text{MS}_S - \text{MS}_E)/m$  when this is nonnegative, where  $\text{MS}_S$  is the between-subject mean square, and that the Wald statistic for  $H : \mu_1 = \cdots = \mu_m$ , divided by  $m - 1$ , is the uncorrected  $F$  of Eq. 33.3.1.

### 33.3.6.3 C. Going deeper

### Exercise. C1

For  $m = 4$ ,  $N = 20$  and  $\Sigma$  compound symmetric, compare by simulation the power of the uncorrected univariate test, the Greenhouse–Geisser test and the multivariate test of Proposition 33.3.4 against a linear trend in the occasion means. Repeat with an autoregressive  $\Sigma$  of the same total variance. Which would you choose if you knew the alternative were a linear trend, and why is a single contrast better still?